19/12/22

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech 3rd Semester Final Examination, December 2022

Subject: Mathematics-III(MA 2101)

(All Engineering Branches)

Time : 3 hours | Full Marks : 50

Answer any five questions

1. a) Use axiomatic definition of probability to show that for any two events A and B:

$(i)\ P(A+B) = P(A)+P(B)-P(AB)$, $(ii)\ P(AB) \leq P(A) \leq P(A+B) \leq P(A)+P(B)$.

b) From 20 light bulbs, 4 of which are burnt out, you select 3 bulbs at random. Find the probability that (i) all are burnt out (ii) none are burnt out (iii) at least one is burnt out (iv) exactly one is burnt out.

[(2+3)+5]

2. a) State and prove Tchebycheff inequality for a continuous random variable. Using Tchebycheff's inequality, find a lower bound for the probability of getting 64 to 184 driving licences issued by Road Transport Authority in a specific month. It is given that the number of driving licences issued per month be a random variable having mean $m = 124$ and standard deviation $\sigma = 7.5$.

b) Show that the correlation coefficient between two random variables lies between -1 and +1 both inclusive.

c) In a partially destroyed Laboratory record of an analysis of correlation data, the following results only are legible: Var(X)=9.

Regression equations: $8X - 10Y + 66 = 0,\ 40X - 18Y = 214$.

Find (i) the mean values of X and Y, (ii) $\rho(X,Y)$, (iii) σ_Y.

[(1+2+2)+2+(1+1+1)]

3. a) Find $L^{-1}\left\{\frac{4}{2s-3} - \left(\frac{3+4s}{9s^2-16}\right) + \frac{8-6s}{16s^2+9}\right\}$.

b) Using Convolution theorem, find the inverse Laplace transform of

$$\frac{s^2}{(s^2+a^2)(s^2+b^2)},\ a^2 \neq b^2.$$

[5+5]

4. a) Find the mean and variance of Binomial distribution.

b) It is observed that a cricket player becomes out within 10 runs in 3 out of 10 innings. If he played 4 innings, what is the probability that he will become (i) out twice (ii) out at least once within 10 runs?

c) The length of bolts produced by a machine is normally distributed with mean 4 and standard deviation 0.5. A bolt is defective if its length does not lie in the interval (3.8, 4.3). Find the percentage of defective bolts produced by the machine. Given:

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.6} e^{-t^2/2}dt = 0.7257, \quad \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.4} e^{-t^2/2}dt = 0.6554.$$

[3+3+4]

5. a) The probability density function of a two-dimensional random variable (X, Y) is given by

$$f(x,y) = \begin{cases} k(x+y), & x \geq 0,\ y \geq 0,\ x+y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find k and evaluate $P(X < \frac{1}{2},\ Y > \frac{1}{4})$.

b) Consider the set of equations

$$\begin{aligned} 4x_1 + 2x_2 - 3x_3 &= 1 \\ 6x_1 + 4x_2 - 5x_3 &= 1 \end{aligned}$$

Verify that $(2, 1, 3)$ is a feasible solution. Reduce this feasible solution to a basic feasible solution. [5+ 5]

6. a) Define hyperplane. Show that a hyperplane is a convex set.

b) Use simplex method to solve the following L.P.P.

$$\begin{aligned} \text{Max } z &= 7x_1 + 5x_2 \\ \text{Subject to } x_1 + 2x_2 &\leq 6 \\ 4x_1 + 3x_2 &< 12 \\ x_1 \geq 0,\ x_2 &\geq 0. \end{aligned}$$

[(1+2)+ 7]

7. a) Find all the basic solutions to the system of linear equations

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 2 \\ 3x_1 + 2x_2 + x_3 &= 3. \end{aligned}$$

b) Use Big-M method to solve

$$\begin{aligned} \text{Max } z &= 3x_1 + 2x_2 + 3x_3 \\ \text{Subject to } 2x_1 + x_2 + x_3 &\leq 2 \\ 3x_1 + 4x_2 + 2x_3 &\geq 8 \\ x_i \geq 0, i &= 1, 2, 3. \end{aligned}$$

[3+7]